# Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (`uXXXXXXX@anu.edu.au`); PE supports ANU SSO.

   https://pollev.com/anuiml2026

2. Keep the webpage open. When the lecturer announces a quiz question, the webpage will be updated accordingly.

# Science and Inductive Reasoning

# Lecture 1 - Quiz 1 (120 seconds)

**L1Q1** - Which of the following correctly describes the relationship between inductive and deductive reasoning, as discussed in the lecture?

- A) They are mutually exclusive; a scientist must commit to one or the other.
- B) Induction is logically safe, while deduction can be overturned by a single counter-example.
- C) Deduction is logically safe, meaning true premises guarantee a true conclusion, but induction cannot logically guarantee a universal claim from finitely many observations.
- D) Inductive reasoning always precedes deductive reasoning in the history of science.

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# Lecture 1 - Quiz 2 (120 seconds)

**L1Q2** - The lecture draws an analogy between a hypothesis set that can perfectly fit *any* possible dataset and the "secret society" example. What is the shared problem?

- A) Both require too many parameters to be computationally tractable.
- B) Both are unfalsifiable since they accommodate every possible observation and therefore explain nothing.
- C) Both lead to slow convergence of the learning algorithm.
- D) Both violate the assumption that training data is representative.

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# Types of Machine Learning

# Lecture 1 - Quiz 3 (120 seconds)

**L1Q3** - A company wants to compress gene-expression data (thousands of genes per patient) into a small number of variables that capture the main sources of variation. What ML type and task does this correspond to?

- A) Supervised learning—regression.
- B) Unsupervised learning—dimensionality reduction.
- C) Reinforcement learning—policy learning.
- D) Supervised learning—classification.

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# Lecture 1 - Quiz 4 (120 seconds)

**L1Q4** - The lecture states that all three types of ML share the same underlying inductive challenge. What is it?

- A) Defining a reward signal that correctly grades every possible action.
- B) Ensuring labels are collected without measurement noise.
- C) Generalising beyond what has been observed in the data.
- D) Choosing a hypothesis set that is neither too simple nor too complex.

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# The (Supervised) Learning Framework

# Lecture 1 - Quiz 5 (90 seconds)

**L1Q5** - The in-sample error is defined as

$$E_{\text{in}}(h) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\{h(\mathbf{x}_i) \neq y_i\}$$

If a hypothesis misclassifies exactly 3 out of 10 training examples, what is $E_{\text{in}}(h)$?

- A) 3
- B) 7
- C) 0.7
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# Lecture 1 - Quiz 6 (90 seconds)

**L1Q6** - What is the risk of choosing a hypothesis set $\mathcal{H}$ that is *too simple*?

- A) The model becomes unfalsifiable and cannot be tested.
- B) The learning algorithm will overfit the training data.
- C) No hypothesis in $\mathcal{H}$ may be close to the true target $f$, regardless of how much data is collected; this is called the *approximation error*.
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# A Concrete Example: The Perceptron

# Lecture 1 - Quiz 7 (90 seconds)

**L1Q7** - What does the parameter $w_0$ (the bias) control in the perceptron in the credit approval example?

- A) The orientation of the decision boundary.
- B) The scale of the feature weights.
- C) Where the approve/deny threshold is located, independently of the features.
- D) The learning rate of the PLA.

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# Lecture 1 - Quiz 8 (90 seconds)

**L1Q8** - What does the convergence theorem for the PLA guarantee, and under what condition?

- A) The PLA always finds the unique optimal linear boundary, regardless of the data.
- B) If the data is linearly separable, the PLA stops after finitely many updates with zero training error; if not, it loops forever.
- C) The PLA converges to a solution with minimum in-sample error even when the data is not linearly separable.
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